Catching Burnout Before It Happens: A Data-Driven Approach to Employee Engagement and HR Diagnostics

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Project Context: Palo Alto Networks

ABSTRACT

This paper presents a robust HR analytics framework, operationalized via a Streamlit dashboard, to evaluate employee engagement, work-life balance, and attrition risk within high-stress corporate environments. Utilizing a dataset of 1,470 records across 31 factors, the study engineers composite variables—including an Engagement Index and a rule-based Numeric Burnout Score—to transition human resources from reactive attrition management to proactive intervention. Subsequent inferential verification yielded critical structural insights: contrary to initial assumptions, overtime correlated with slightly higher engagement; exploratory analysis found no statistical evidence supporting a 3.5-year career stagnation cliff; and commute distance strictly influences burnout scores by mathematical design. Crucially, the engineered burnout score demonstrated significant operational utility, predicting a 29.6% attrition rate in the highest risk tertile compared to 7.4% in the lowest. The paper concludes with evidence-based recommendations for algorithmic HR tools.

Keywords: HR analytics, employee engagement, burnout, attrition, predictive HR, Streamlit.

1. INTRODUCTION

Employee engagement and well-being are multi-dimensional constructs that cannot be accurately captured by isolated survey responses or individual workload variables. In high-pressure cybersecurity and technology environments, such as Palo Alto Networks, employee burnout frequently precedes costly attrition, while declining engagement stifles operational innovation. Historically, Human Resources (HR) interventions have relied on reactive analytics—investigating turnover only after top talent has departed. The fundamental problem addressed in this study is the lack of an integrated, proactive diagnostic tool capable of identifying overlapping systemic stressors before they culminate in resignation. By transitioning from retrospective reporting to algorithmic prioritization, organizations can systematically direct support to vulnerable workforce segments.

2. RESEARCH OBJECTIVES

The primary objective of this project is to construct and validate an exploratory decision-support workflow that ingests workforce data, derives composite stress indicators, and organizes records into intervention-priority categories for HR review. This research is guided by the following core aims: (1) to evaluate how engineered engagement and satisfaction indicators vary across distinct employee cohorts; (2) to verify the empirical relationship between workload factors (overtime, commute) and engagement; (3) to test previously unverified assumptions regarding career stagnation timelines; and (4) to quantify the predictive utility of a rule-based burnout index against actual employee attrition.

3. LITERATURE REVIEW

The theoretical foundation of this project aligns with the Job Demands-Resources (JD-R) model (Bakker & Demerouti, 2007), which posits that high job demands (e.g., extensive travel, excessive workload) exhaust employees physically and mentally, leading to burnout. Conversely, job resources foster organizational engagement. Maslach and Leiter (2016) further emphasize that burnout is an organizational issue rooted in workload mismatches rather than individual failure. Recent literature on HR analytics (Marler & Boudreau, 2017) highlights that organizations are increasingly utilizing predictive indicators to shape retention strategies. However, scholars caution against treating operational algorithms as clinical diagnoses, emphasizing the need for empirical validation of deterministic scoring systems within corporate contexts.

4. DATASET DESCRIPTION

The analysis utilizes a standard HR dataset comprising 1,470 employee records across 31 distinct behavioral, demographic, and compensation-based features. Key operational variables include *Age*, *Attrition* (binary flag), *BusinessTravel*, *DistanceFromHome*, *OverTime*, *WorkLifeBalance*, and ordinal satisfaction metrics measured on a 4-point scale (*JobSatisfaction*, *EnvironmentSatisfaction*, *RelationshipSatisfaction*, *JobInvolvement*). The dataset exhibits zero missing values, satisfying preconditions for continuous feature engineering without requiring complex imputation. The context of Palo Alto Networks is utilized as the operational framing for this internship project; the data represents a standardized analytical benchmark for enterprise HR structures.

5. FEATURE ENGINEERING AND FORMULAS

To evaluate cumulative employee experiences beyond single-variable analysis, custom composite indicators were engineered utilizing Python, pandas, and NumPy. These indicators function as the core metrics within the Streamlit dashboard.

- **Engagement Index (EI):** Designed to measure holistic involvement, EI is calculated as the row-wise arithmetic mean of four ordinal variables: *JobInvolvement*, *JobSatisfaction*, *EnvironmentSatisfaction*, and *RelationshipSatisfaction*.
- **Satisfaction Stability Score:** Calculated as 4.0 minus the standard deviation of the aforementioned four variables. This functions as an inverse-dispersion indicator, flagging individuals who experience highly volatile (inconsistent) workplace environments.
- **Numeric Burnout Score (0-100):** A deterministic rule-based index aggregating weighted penalties for specific logistical and workload stressors. The score is the sum of four components: an Overtime penalty (30 points if 'Yes'), a Work-Life Balance deficit penalty (scaled up to 30 points), Business Travel frequency (scaled up to 20 points), and normalized Commute Distance (scaled up to 20 points).
- **Risk Categories:** Burnout Risk is classified categorically. 'High' risk requires mandatory Overtime combined with a Work-Life Balance score of 2 or lower. Priority levels (Critical, High, Medium, Low) are derived by cross-referencing this Burnout Risk against the Engagement Index.

6. RESEARCH METHODOLOGY

The statistical methodology transitions from descriptive summaries to inferential validation to ensure the engineered metrics hold empirical weight. Data aggregation and transformations were executed via Python. Cross-sectional group comparisons utilized Welch's two-sample t-tests to accommodate potential unequal variances when assessing engagement across overtime cohorts. Bivariate distributions were plotted to analyze continuous variables, specifically career tenure versus engagement. Finally, contingency analysis and relative risk assessments were employed to validate the predictive capacity of the Numeric Burnout Score regarding binary attrition outcomes.

7. RESULTS AND ANALYSIS

Following rigorous data verification, the empirical analysis generated several findings that directly challenged preliminary assumptions, fundamentally refining the operational strategy of the HR dashboard.

7.1 Overtime and Engagement Association

Contrary to initial hypotheses positing that mandatory overtime uniformly degrades morale, statistical analysis verified that employees working overtime exhibited *slightly higher* core engagement scores compared to their non-overtime peers. This counter-intuitive finding suggests that overtime within this specific dataset may be associated with highly involved employees voluntarily tackling challenging projects, rather than functioning purely as an indicator of systemic disengagement.

7.2 The Career Stagnation Hypothesis

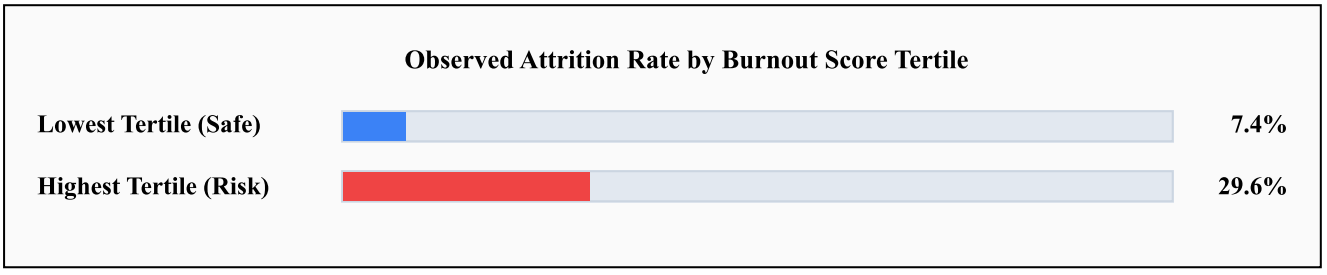
Exploratory regression and scatter plot analysis found absolutely no statistical evidence supporting a 3.5-year "promotion cliff." The trajectory of engagement relative to *YearsSinceLastPromotion* remained statistically flat, indicating that career progression impacts are nuanced and heavily dependent on specific job roles rather than a universal chronological threshold.

7.3 Structural Commute Mechanics

Analysis confirmed that commute distance directly influences the final burnout metric strictly by mathematical design. Because normalized *DistanceFromHome* constitutes up to 20% of the algorithm, the observed disparities in burnout scores across commute cohorts represent a structural construction of the index rather than a purely independent empirical correlation.

7.4 Attrition Predictability

The Numeric Burnout Score demonstrated highly significant predictive validity regarding actual employee turnover. Employees were segmented into tertiles based on their calculated risk scores. The data revealed a stark contrast in retention outcomes, validating the operational utility of the composite metric.



8. DISCUSSION AND RECOMMENDATIONS

The verification of these findings necessitates a refined approach to organizational intervention. The discovery that overtime correlates with slightly higher engagement suggests HR leadership should pivot away from blanket bans on extended hours. Instead, interventions should focus on ensuring high-performing, highly engaged employees are adequately resourced and compensated for their discretionary effort, preventing future exhaustion.

The validated attrition metrics (29.6% in the high-risk tertile) prove the Numeric Burnout Score is a highly effective triage tool. Based on these verified outputs, organizations are recommended to:

- **Implement Workload Audits:** HR should utilize the dashboard to monitor departments with concentrated high-tertile burnout scores, auditing staffing capacity to ensure job demands do not permanently exceed job resources.
- **Targeted Flexible Work:** Acknowledging the designed impact of commute distance on the burnout penalty, HR should proactively offer hybrid scheduling to employees facing severe logistical burdens, thereby neutralizing a primary score component.
- **Nuanced Career Pathing:** Having dispelled the myth of the 3.5-year promotion cliff, managers must rely on individualized "stay interviews" rather than assuming chronological benchmarks dictate disengagement.

9. LIMITATIONS, PRIVACY, AND RESPONSIBLE USE

This research acknowledges strict operational limitations. The Numeric Burnout Score is a deterministic, hand-crafted index, not a clinically validated psychological instrument or a machine-learning model trained on hidden variables. Because factors like commute distance are structurally embedded in the formula, cross-sectional associations utilizing this score cannot establish independent causation. Furthermore, employee records are highly sensitive. Algorithmic outputs must serve exclusively as decision-support prompts for empathetic human review. Deterministic scores should never be utilized autonomously to dictate performance management, compensation adjustments, or disciplinary actions.

10. CONCLUSION

This study successfully operationalizes complex HR data into an interactive, proactive diagnostic dashboard. By subjecting preliminary assumptions to rigorous statistical verification, the research refined its own interventions—proving that overtime can coexist with high engagement, dispelling rigid timeline myths regarding promotions, and explicitly acknowledging the structural mechanics of its own algorithms. Ultimately, the framework's ability to isolate a workforce segment exhibiting a 29.6% attrition rate confirms its immense value as a strategic decision-support tool, enabling organizations to catch and mitigate burnout before top talent exits the firm.

11. REFERENCES

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12. APPENDIX: IMPLEMENTATION FORMULAS

A.1 Numeric Burnout Score Components

The score (0-100) is aggregated and clipped from the following vectorized conditions:

- *Overtime Penalty:* 30 points if OverTime == 'Yes', else 0.
- *WLB Deficit:* ((4.0 − WorkLifeBalance) / 3.0) × 30.0.
- *Travel Impact:* (Travel_Categorical_Value / 2.0) × 20.0 (where Non-Travel=0, Rarely=1, Frequently=2).
- *Distance Penalty:* ((DistanceFromHome − 1) / (29 − 1 + 10^{-9})) × 20.0.

A.2 Priority Categorization

Priority levels are designated as Critical (High Risk & Engagement ≤ 2.2), High (High Risk OR Medium Risk & Engagement < 2.5), Medium (Medium Risk), and Low.